P	Q
T	T
T	F
F	T
F	F

T, F → interpretation.

$(P \cup Q)' = P' \cap Q'$

$(P \cap Q)' = P' \cup Q'$

19-9-22

P	Q	$P \cap Q$	$P \cup Q$	$P \to Q$	$P \leftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

$P \vee (P \wedge Q)$
T
T
F
F

Q. $(\neg A) \vee ((\neg B) \cap C)$

Sol^n

A	B	C	$\neg A$	$\neg B$	$\neg B \cap C$	$(\neg A) \vee ((\neg B) \cap C)$
T	T	T	F	F	F	F
T	T	F	F	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	F	F
F	T	T	T	F	F	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	F	T

FOPL (First Order Predicate Logic)

21-09-22

① All students are smart.

→ $\forall x : [\text{student}(x) \rightarrow \text{smart}(x)]$

② There is a student who is smart.

→ $\exists x : [\text{student}(x) \wedge \text{smart}(x)]$

③ All kings are person.

→ $\forall x : [\text{king}(x) \rightarrow \text{person}(x)]$

④ Everybody loves somebody

→ $\forall x \, \exists y : \text{loves}(x, y)$

not specific ← $\begin{cases} x : \text{Everybody} \\ y : \text{Somebody} \end{cases}$

⑤ Every gardener likes the sun.

→ $\forall x : \text{gardener}(x) \rightarrow \text{likes}(x, \text{sun})$

or

$\forall x : \text{likes}(\text{gardener}(x), \text{sun})$

⑥ All purple mushrooms are poisonous.

→ $\forall x : (\text{Mushrooms}(x) \wedge \text{purple}(x)) \rightarrow \text{poisonous}(x)$

Rules for FOPL :—

22-09-22

① A term (denoting real world object) is a constant symbol, a variable symbol, or a function.

② An atom (which has value true or false) is either an n-place predicate of n terms. If P and Q are atoms then ~~negat~~ $\sim P$, $P \vee Q$, $P \wedge Q$, $P \rightarrow Q$, $P \leftrightarrow Q$ are atoms.

③ A sentence is an atom, or if P is a sentence and x is a variable then $(\forall x) P$ and $(\exists x) P$ are sentences.

④ A well formed formula (WFF) is a sentence containing no free variable i.e. all the variables are bound by universal or existential quantifiers.

For example – $(\forall x) P(x, y)$ has a 'x' bound as a universally quantified variable but 'y' is free. So it is not a WFF.

WFF's are constructed using the following rules —

① True and false are WFF

② Each propositional constant and each propositional variables are WFF.

③ Each atomic formula is a WFF.

④ If A, B, C are WFF then $\sim A$, $A \vee B$, $A \cap B$, $A \rightarrow B$, $A \leftrightarrow B$ are also ~~are~~ WFF.

⑤ If x is a variable and A is a WFF then $(\forall x) A$ and $(\exists x) A$ are also WFF.

$\forall : \rightarrow$
$\exists : \wedge$

① Tommy is a Dog.

→ Dog (Tommy).

② The Dog has a tail.

→ Has-a-tail (Dog)

③ All dogs have tail

~~Have-tail (∀ dog)~~

→ $\forall x : (Dog(x) \rightarrow has\text{-}a\text{-}tail(x))$

④ All persons need air

→ $\forall x : (person(x) \rightarrow need\text{-}air(x))$

⑤ The owner of the car also owns the truck

→ $\exists x :$ owns $(x$, car$)$ $\wedge$ owns $(x$, truck$)$

⑥ All Romans were either loyal to the caesar or hated him.

→ $\forall x : [Roman(x) \longrightarrow Loyal\text{-}to(x, caesar) \vee hated(x, caesar)]$

⑦ Marcus is a man.

→ Man (Marcus)

⑧ Marcus was a man

→ Man (Marcus)

Drawback of FOPL
→ we can't define the tense

(7) All pompeians are Roman.

→ $\forall x$: pompeian $(x) \rightarrow$ Roman (x)

(8) Caesar was a ruler.

→ Ruler (caesar)

(9) Everyone is loyal to someone.

→ $\forall x \exists y$: Loyal (x, y)

(10) Someone is loyal to someone.

→ $\exists x \exists y$: loyal (x, y)

(11) Everyone is loyal to everyone.

→ $\forall x \forall y$: loyal (x, y)

(12) An equilateral triangle has all sides and angle equivalent

→ $\forall x$: equilateral triangle $(x) \rightarrow$ equivalent side $(x) \wedge$ equivalent angle (x).

Resolution :—

Resolution produces proofs by refutation. In other words to prove a statement (i.e. to show that it is valid). resolution attempts to show that the negation of the statement produces a contradiction with the known statement (i.e. that it is unsatisfyable)

Conversion to clause form —

Algorithm ~~ass~~

① Eliminate → using the fact that $a \to b$ is equivalent to $\sim a \vee b$. $P \leftrightarrow Q \cong (\sim P \vee Q) \wedge (\sim Q \vee P)$

② Reduce the scope of each negation to a single term, using the fact that $\sim(\sim P) \cong P$,

De Morgan's Laws ~~$\sim A \wedge B$~~

$$\sim(a \wedge b) = (\sim a) \vee (\sim b)$$

and $\sim(a \vee b) = (\sim a) \wedge (\sim b)$,

The standard correspondence between quantifiers

$$\sim \forall x : P(x) = \exists x : \sim P(x)$$
$$\sim \exists (x) : P(x) = \forall x : \sim P(x)$$

③ Standardized variables show that every quantifier binds a singular variable

④ Move all quantifiers to the left of the formula while not dynamical their relative order this is attainable since there is no conflict among variable names.

(5) Eliminate existential quantifiers ($\exists$). 26-09-22

(6) Drop the prefix. At this time all remaining variables are universally quantified, therefore the prefix will simply be been and any ~~rem~~ proof procedure we have a tendency to use will nearly assume that any variable it sees is universally quantified.

(7) Convert the matrix into a conjunction of disjunctions.

(8) produce a separate clause reminiscent of every conjunction

Eg:- $\exists x \forall y (\forall z\ P(f(x), y, z) \rightarrow (\exists u\ Q(x,u)\ \&\ \exists v\ R(y,v)))$.

Convert this equation into equivalent clause form.

$$\exists x \forall y (\sim(\forall z) P(f(x), y, z) \vee (\exists u\ Q(x,u) \wedge \exists v\ R(y,v)))$$

$$\Rightarrow \exists x \forall y (\exists z \sim P(f(x), y, z) \vee (\exists u\ Q(x,u) \wedge \exists v\ R(y,v)))$$

$$\Rightarrow \forall y (\sim P(f(a), y, g(y)) \vee (Q(a, h(y)) \wedge R(y, k(y)))$$

$$\Rightarrow \forall y (\sim P(f(a), y, g(y)) \vee Q(a, h(y))) \wedge (\sim P(f(a), y, g(y)) \vee R(y, k(y)))$$

So the clauses are -

$$\sim P(f(a), y, g(y)) \vee Q(a, h(y))$$

$$\sim P(f(a), y, g(y)) \vee R(y, k(y))$$

Skolemization :-

Skolemization, the replacement of existentially quantified variables with skolem functions and deletion of the respective quantifiers, is then accomplished as follows -

If the first ~~quantifier~~ (left most) quantifier in an expression is an existential quantifier, replace all occurances of the variable it quantifies with an arbitrary constant not appearing else where in the expression and delete the quantifier.

This same procedure should be followed for all other existential quantifiers not preceded by a universal quantifier, using different constants symbols in the substitution.

For each existential ~~ex~~ quantifier i.e. preceded by one or more universal quantifiers, replace all occurances of the existentially quantified variables by a function symbol not appearing else where in the expression. The arguements assigned to the function should match all the variables appearing in each universal quantifier which precedes the existential quantifier. This existential ~~ex~~ quantifier should then be deleted. The same process should be repeated for each remaining existential quantifier using a different function symbol and choosing function arguments that corresponds to all universally quantified variables

that precede the existentially quantified variable being replaced.

22-09-22

Q. Consider the following statements

i) If the maid stole the jewellery, then the butler was not guilty.

ii) Either the maid stole the jewelery, or she milked the the cow.

iii) If the maid milked the cow, then the butler got the cream.

iv) Therefore, if the butler was guilty then he got his cream.

a) Express in propositional calculus.

b) Express the negation in clausal form.

c) Demonstrate that the conclusion is valid using resolution in propositional calculus.

Soln: i) P = Maid stole the jewellery.

Q = Butler is guilty.

$P \rightarrow \sim Q$.

ii) R = She milked the cow.

$P \vee R$

iii) S = Butler got the cream

$R \rightarrow S$

iv) $Q \rightarrow S$. (conclusion)

b) <u>Convert into CNF</u>

(i) $\sim P \vee \sim Q$

(ii) $P \vee R$

(iii) $R \rightarrow S = \sim R \vee S$

(iv) $Q \rightarrow S = \sim Q \vee S$

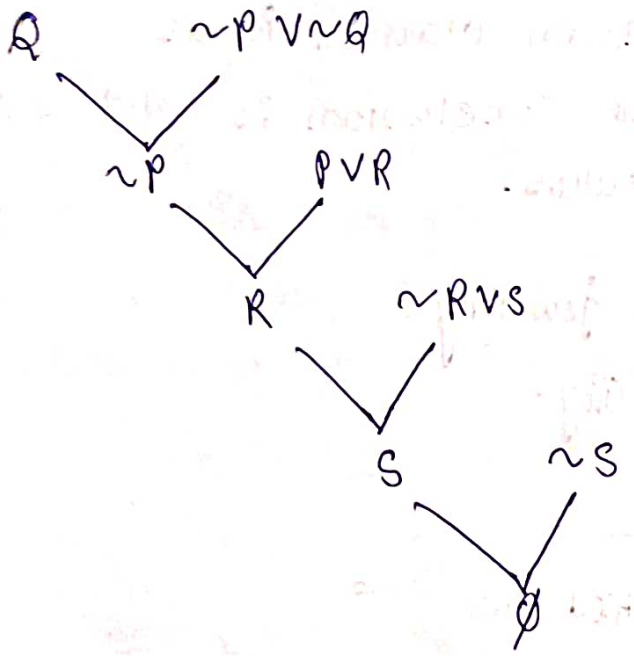
(c) $\sim(\sim Q \vee S) = \sim(\sim Q) \wedge \sim S$

$= Q \wedge \sim S.$

29-09-22

Resolution:-

(Always initiated from the conclusion)



Hence we see that the negation of the conclusion has proved a contradiction with the given set of facts. That means the negation of the conclusion is invalid or false, as we can say that the assertion of the conclusion is valid or true. Hence the required conclusion is proved.

Q. Consider the following information –

(i) Ravi likes all kind of food.
(ii) Fish and chicken are food.
(iii) Anything anyone eats, and is not killed, it is food.
(iv) Ajay eats peanut and is still alive.
(v) Rita eats everything Ajay eats.

prove a) Ravi likes Peanuts using resolution.
b) Answer the question what food does Rita eat using resolution.

Soln:

(i) $\forall x$: Food $(x) \longrightarrow$ Likes(Ravi, x)
(ii) food (fish) $\wedge$ food (chicken).
(iii) $\forall x \forall y$: eats $(x, y) \wedge \sim$ killed $(x) \rightarrow$ food (y)
(iv) ~~Peanut (x)~~ eats (Ajay, peanuts) $\wedge$ alive (Ajay)
~~(iv) $\forall x$: eats (Ajay, x) // eats~~
(v) $\forall z$: eats (Ajay, z) $\rightarrow$ eats (Rita, z).

Converting to CNF

(i) $\sim$food$(x) \vee$ Likes (Ravi, x)
(ii) food (fish) $\wedge$ food (chicken).

food (fish)
food (chicken).

(iii) $\sim$ (eats $(x, y) \wedge \sim$ killed (x)) $\vee$ food (y).

((~ eats (x, y)) $\vee$ killed (x)) $\vee$ food (y).

(iv) eats (Ajay, peanuts)

alive (Ajay)

(v) ~ eats (Ajay, z) ∨ eats (Rita, z).

(a) ~Likes (Ravi, peanuts)

~ food (x) ∨ likes (Ravi, x)

~ food(x)

~eats (x, y) ∨ Killed(x) ∨ food(y)

~eats (x, y) ∨ Killed (x)

eats (Ajay, peanu

Killed (x)

~alive(p) ∨ ~Killed

~ alive (p)

alive(Ajay)

∅

(b) ~~eats (Rita, z) ∨ ~eats (Rita, z)~~

Note: In our predicates we have alive(x) and killed(x), which actually mean the same but in computer the resolution technique will not recognize it. Hence either we replace any one by another or add two facts as follows —

$\forall P$: alive(P) $\rightarrow$ ~ killed(P)

$\forall Q$: ~ killed(Q) $\rightarrow$ alive(Q)

Converting it into CNF

~ alive(P) $\vee$ ~ killed(P)

killed(Q) $\vee$ alive(Q)

~~Consider the following statements~~

1) History of AI
2) predicate logic
3)

(6) eats (Rita, z) ∨ ~ eats (Rita, z) ~ eats (Ajay, z) ∨ eats (Rita, z)

eats (Rita, z) ∨ ~ eats (Ajay, z) eats (Ajay, peanuts)

eats (Rita, peanuts).

---

5. consider the following statements

(i) who ever can read is literate.

(ii) Animals are not literate.

(iii) Some animals are intelligent

prove the following by resolution.

(iv) Some who can not read are also intelligent

Soln: converting the given statements to predicate logic —

(i) $\forall x$: Read (x) → literate (x).

(ii) $\forall y$: Animals (y) → ~ literate (y).

(iii) $\exists y$: animals (y) ∧ Intelligent (y)

(iv) $\exists z$: ~ read (z) ∧ intelligent (z)

converting to CNF

(i) ~ Read (x) ∨ literate (x)

(ii) ~ Animals (y) ∨ ~ literate (y)

*. Neural Network 10/10/22

(i) Single layer
(ii) Multi layer
(iii) Intermediate layer / hidden layer

Q(1) Explain about biological neuron with the help of a diagram in 200 words.

(2) Difference between biological neuron and artificial neuron.

Soln:

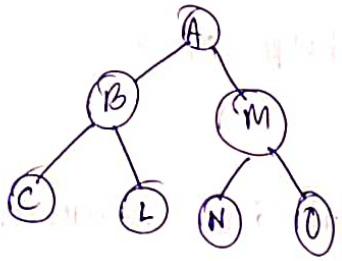
Artificial Neuron	Biological Neuron.
(i) processing speed is fast as compared to biological neuron.	(i) They are slow in processing information.
(ii) Allocation for storage to a new process is strictly irreplaceable as the old location is saved for the previous process.	(ii) Allocation for storage to a new process is easy as it is added just by adjusting the interconnection strengths.
(iii) processes operate in sequential mode.	(iii) The process can operate in massive parallel operations.
(iv) If any information gets corrupted in the memory it cannot be retrieved.	(iv) Information is distributed into the network throughout into sub-nodes, even if it gets corrupted it can be retrieved.
(v) The activities are continuously monitored by a control unit.	(v) There is no control unit to monitor the information being processed into the network.

12-10-22

Searching Algorithm

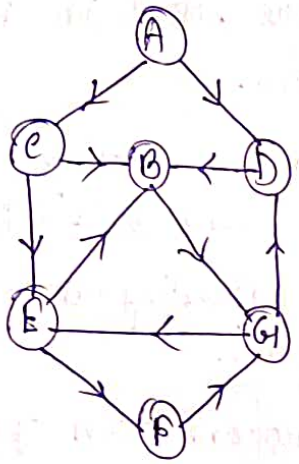
(i) Informed

(ii) Uninformed — DFS (Depth First Search)
— BFS (Breadth First Search)



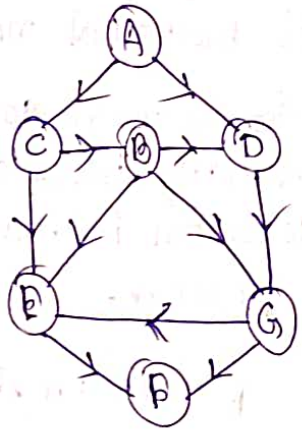
DFS = ABCLMNO

BFS = ABMCLNO → (level wise)



DFS = ACEFGDB

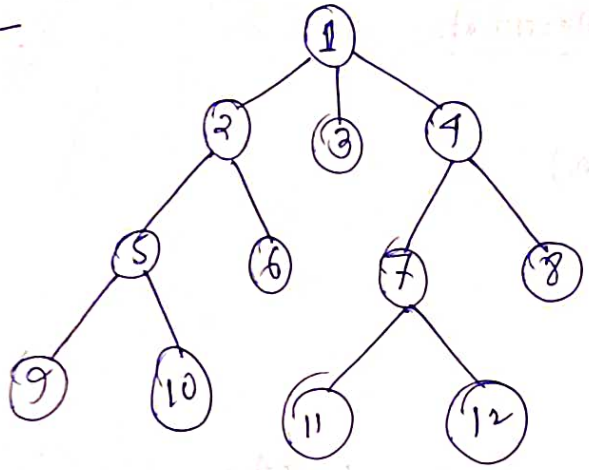
BFS = ACDEBGF



DFS = ACEFBGD

BFS = ACDEBGF

g-

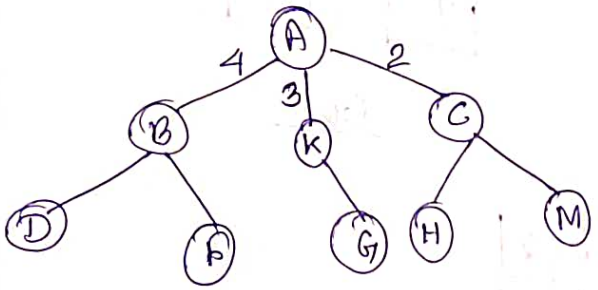


DFS = 1 2 5 9 10 6 3 4 7 11 12 8

BFS = 1 2 3 4 5 6 7 8 9 10 11 12

13-10-22

Best First Search :—



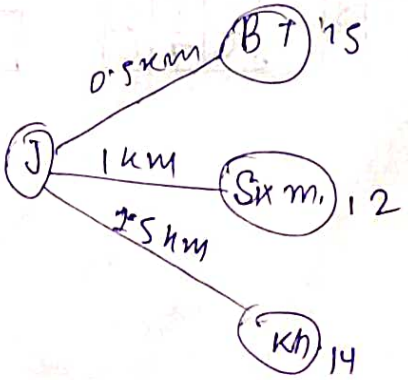
open [ ]
close [ ]

open [A]
close [ ]

open [B K C]
close [A]

*. Heuristic function

$$f(n) = h(n) + g(n)$$



$f(n) = 0.5 + 15 = 15.5$

$1 + 12 = 13$

$1.5 + 14 = 15.5$

$f(n) = g(n)$

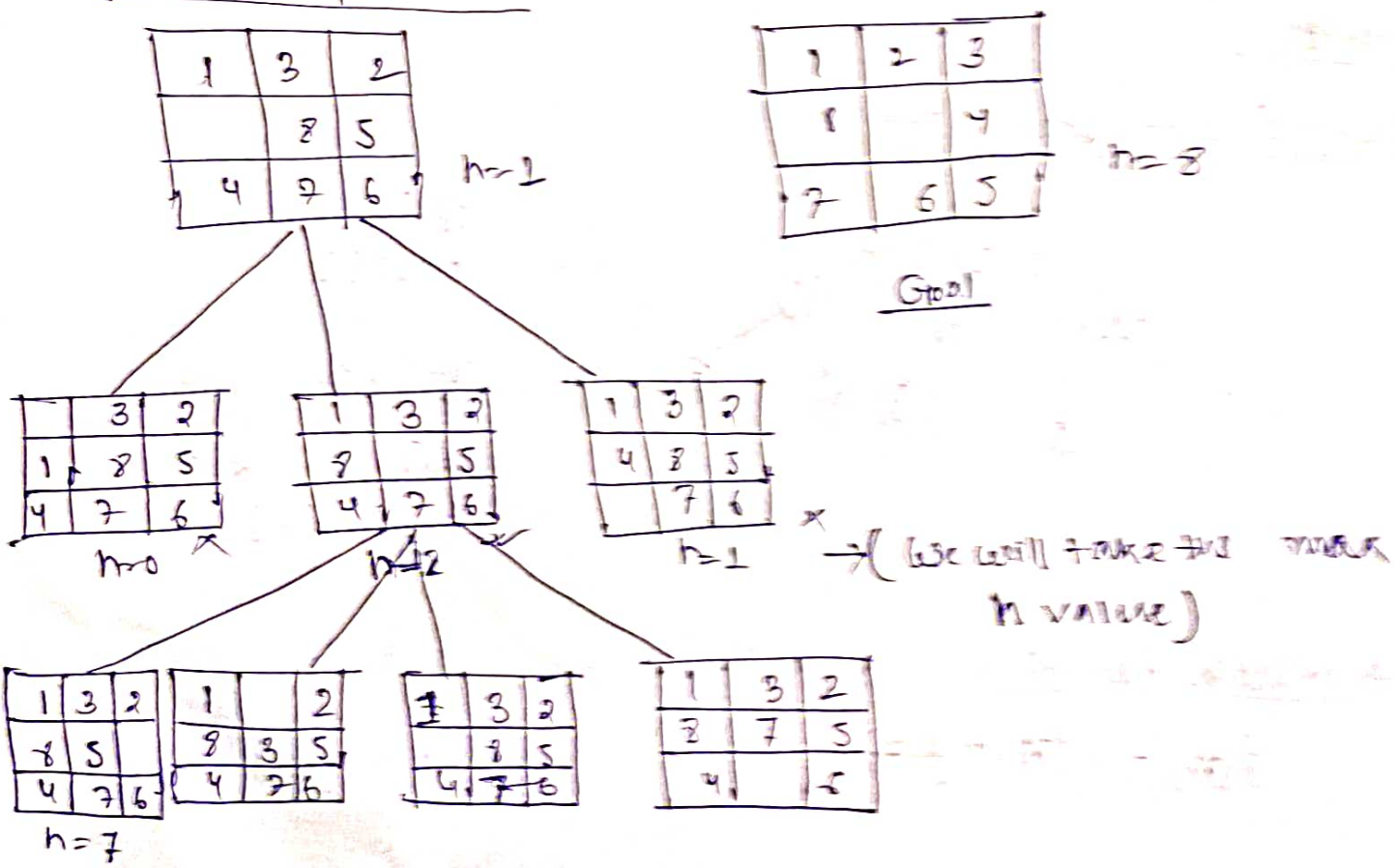
(*) A* Algorithm

Hill climbing problem in Heuristic Search.

- Plateau — (flat area)
- Ridge — (peak)
- foot hill

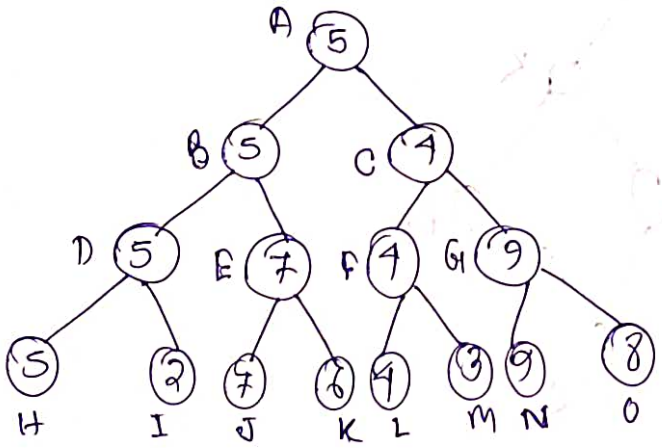
* Statespace → (no. of alternatives used in a searching algorithm).

*. 8-puzzle problem



20-10-22

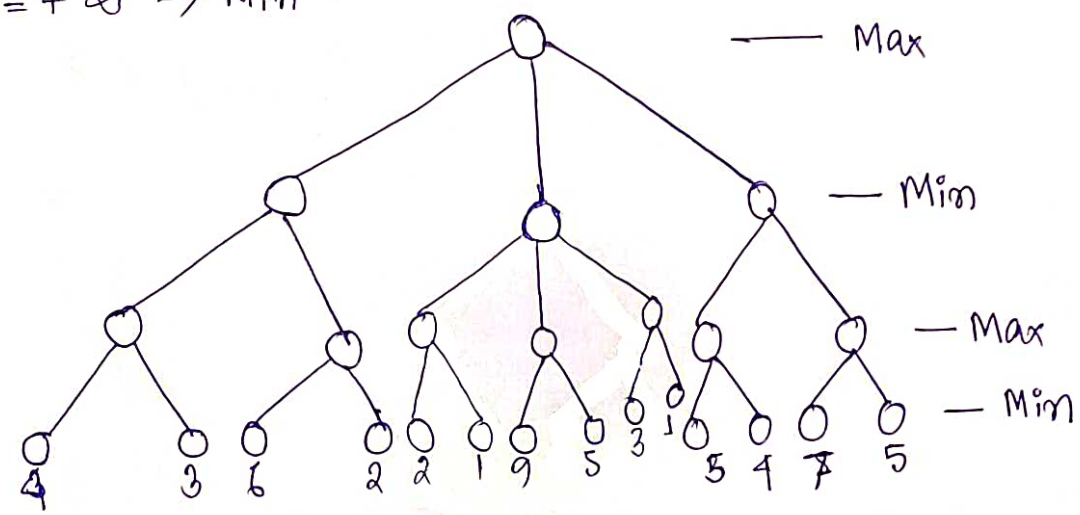
*) Minimax Algorithm :-



*) α - β pruning :-

$\alpha = -\infty \rightarrow$ Max

$\beta = +\infty \rightarrow$ Min



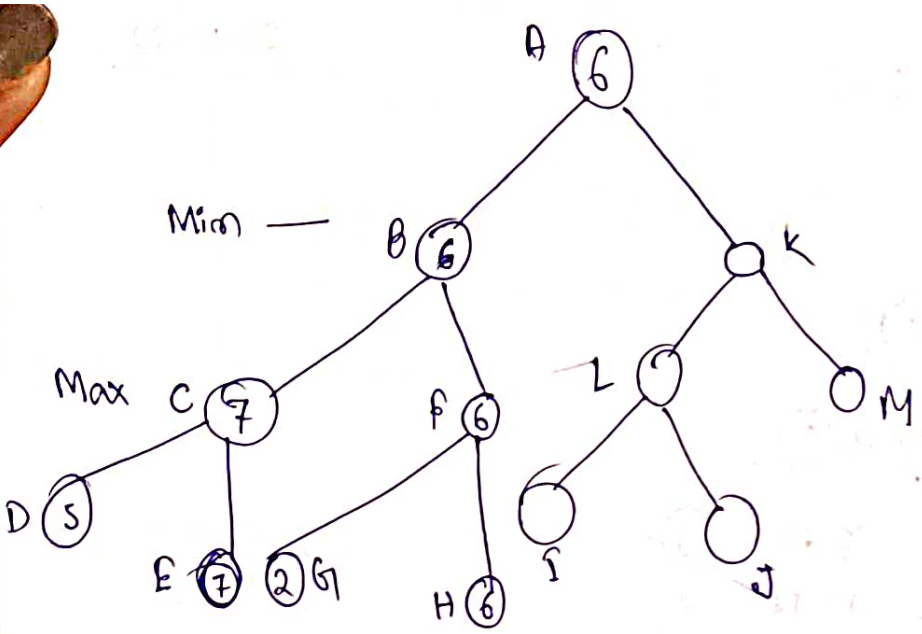
For Max → replace α

For min → replace β.

[if β <= α then no need to go to the right side of the subtree]

*) α-β pruning
*) Minimax algorithm } Theory
(relation)

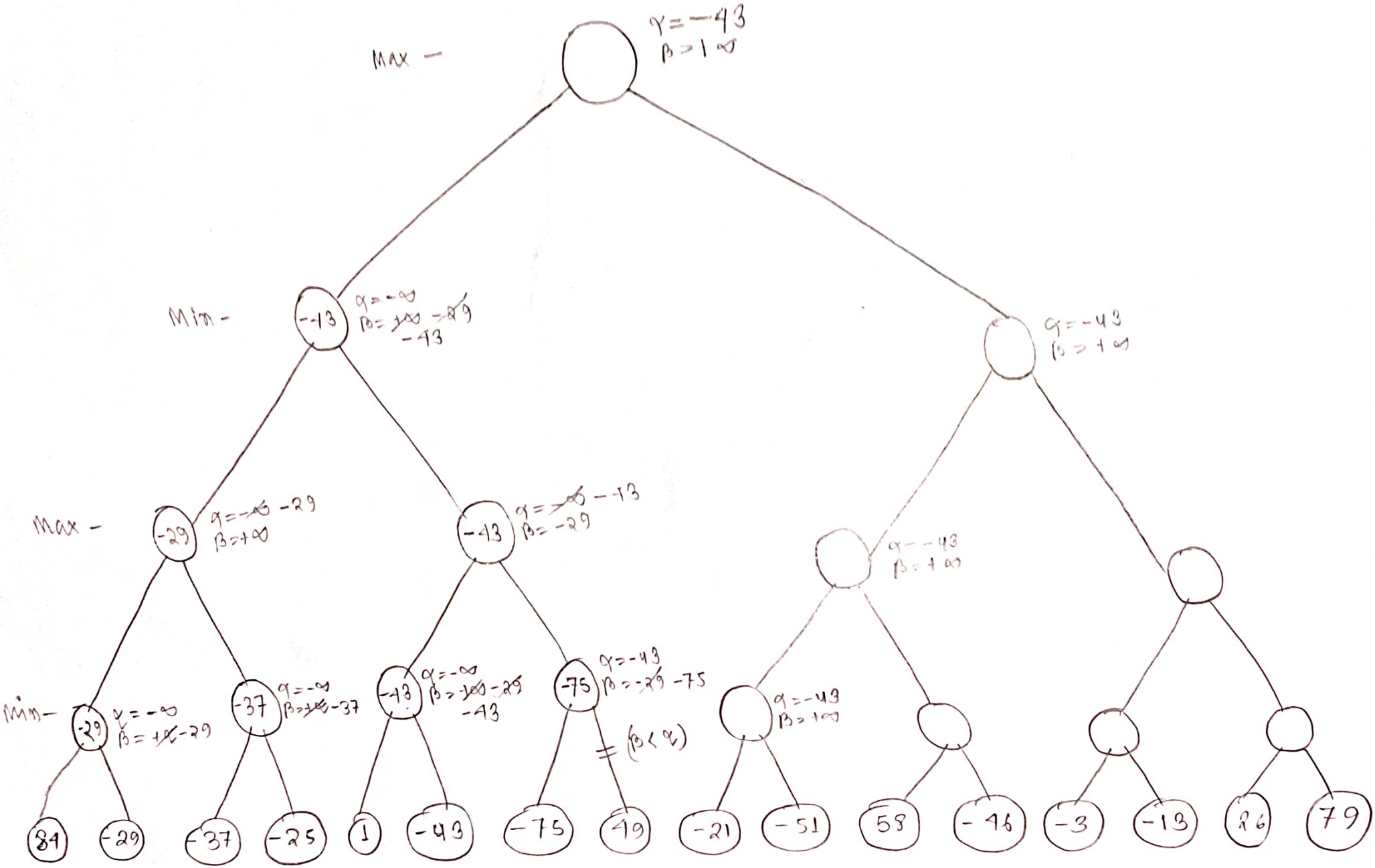
NPTel.



Q.) 84, -29, -37, -25, 1, -43, -75, 49, -21, -51, 58, -46, -3, -13, 26, 79.

Same value

Change value


max -
α = -43
β = +∞
Min -
-13
α = -∞
β = ~~+∞~~ ~~-29~~ -43
α = -43
β = +∞
max -
-29
α = ~~-∞~~ -29
β = +∞
-43
α = ~~-∞~~ -13
β = -29
α = -43
β = +∞
min -
-29
α = -∞
β = ~~+∞~~ -29
-37
α = -∞
β = ~~+∞~~ -37
-43
α = -∞
β = ~~+∞~~ ~~-29~~ -43
-75
α = -43
β = ~~-29~~ -75
= (β < α)
α = -43
β = +∞
84
-29
-37
-25
1
-43
-75
-19
-21
-51
58
-46
-3
-13
26
79